nature portfolio

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Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☒	☐	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☐	☒	A description of all covariates tested
☐	☒	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	All tools used for data collection are publicly available. All sequence data was collected from publicly available portals (UniProt, and JGI). Sequence redundancy reduction was conducted using CD-hit (v4.8.1) and sequence searches were conducted with HMMER (v3.3.2).
Data analysis	All tools used for analyses are publicly available (established tools or previously published). Multiple sequence alignments were conducted with Clustal Omega (v1.2.2). Sequence similarity networks were generated with BLAST (v2.9.0) and analyzed with Cytoscape (v3.8) or Meta-SSN (v1.0.1, https://github.com/johnchen93/MetaSSN). DCA was conducted with the adabmDCA (v1.0) package. Structural data was processed using PyMol (v2.5.5), and multiple structural alignments were performed with the mTM-align webserver (https://yanglab.nankai.edu.cn/mTM-align/). Deep sequencing data was processing using DMS-FastQ-processing (v1.0.1, https://github.com/johnchen93/DMS-FastQ-processing). All other analyses were done with standard Python or Julia libraries.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

All data relevant to the main findings are supplied as source data or supplementary data files. The raw deep sequencing data is available on NCBI SRA under BioProject PRJNA974578.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	Not applicable.
Reporting on race, ethnicity, or other socially relevant groupings	Not applicable.
Population characteristics	Not applicable.
Recruitment	Not applicable.
Ethics oversight	Not applicable.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	No sample size selection was necessary. The experiments and computations were conducted on the entire library of variants, or collected set of sequence data.
Data exclusions	Deep sequencing data was filtered by quality scores and expected chance of variants due to sequencing errors to exclude data that would contribute error or noise. Sequence data for DCA was filtered to only include sequences with the correct active-site motif, and to exclude sequences with too many gaps in the multiple sequence alignment. See methods for details.
Replication	Deep mutational scanning experiments were conducted in duplicate on different days. All experiments were reproducible, as shown by the strong replicate correlations in the supplementary figure.
Randomization	No randomization was necessary, as the full library of variants were tested at once.
Blinding	No blinding was necessary, as the full library of variants were tested at once.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☐	☒ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Dual use research of concern

Policy information about dual use research of concern

Hazards

Could the accidental, deliberate or reckless misuse of agents or technologies generated in the work, or the application of information presented in the manuscript, pose a threat to:

No	Yes
☒	☐ Public health
☒	☐ National security
☒	☐ Crops and/or livestock
☒	☐ Ecosystems
☒	☐ Any other significant area

Experiments of concern

Does the work involve any of these experiments of concern:

No	Yes
☒	☐ Demonstrate how to render a vaccine ineffective
☐	☒ Confer resistance to therapeutically useful antibiotics or antiviral agents
☒	☐ Enhance the virulence of a pathogen or render a nonpathogen virulent
☒	☐ Increase transmissibility of a pathogen
☒	☐ Alter the host range of a pathogen
☒	☐ Enable evasion of diagnostic/detection modalities
☒	☐ Enable the weaponization of a biological agent or toxin
☒	☐ Any other potentially harmful combination of experiments and agents

Precautions and benefits

Biosecurity precautions	All variants generated (including those with higher antibiotic resistance) were only tested in the UBC lab, inside non-pathogenic E. coli lab strains. All cultures grown in the experiments were treated with >10% bleach to destroy living cells and plasmids.
Biosecurity oversight	The experiments were conducted at UBC, with oversight from UBC risk management, as well as the building's safety committee.
Benefits	A more general understanding of antibiotic resistance proteins could help in design of new drugs and in mitigation of resistance. The understanding of protein epistasis could be widely applicable in sequence function prediction and protein engineering.
Communication benefits	The benefits of communication greatly outweighs the risk, as there is useful knowledge to be disseminated to a variety of fields, ranging from evolutionary biology to protein engineering. In addition, most variants in this study confer reduced resistance, and those with higher resistance are unlikely to be useful or easy to employ for malicious means.

Plants

Seed stocks

Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.

Novel plant genotypes

Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.

Authentication

Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosiacism, off-target gene editing) were examined.